

Computational vs. Experimental Determined Spectra of Indigo Dye

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INTRODUCTION

Quantum mechanics aims to provide a theoretical and mathematical framework to describe the behavior of molecules and atoms. A large hurdle in computational chemistry is that describing exact solutions of the Schrödinger equation is extremely demanding, scaling with the complexity of the molecular system. The rise of *Density Functional Theory* (DFT) simplified these calculations by approximating electronic energies based on electron density, rather than multiple different wavefunctions that all interact simultaneously. Walter Kohn and John Pople both received the Nobel Prize in Chemistry in 1998 for their contributions to the field. Kohn received his for the theoretical framework of DFT, while Pople received his for the development of a practical computational method and the creation of the software behind it.

Computational chemistry heavily relies on DFT and *Time-Dependent Density Functional Theory* (TDDFT) to simulate the ground and excited states, which are present in absorption spectra. These predictions should align with the experimentally observed spectra in the UV-Visible region.

Computational chemistry techniques were utilized in this experiment to determine the absorption spectrum of *Indigo* ($C_{16}H_{10}N_2O_2$). Indigo is a historically and industrially significant dye. It is responsible for the blue color of denim and is one of the most widely produced organic dyes in the modern world.

Due to the deep color of indigo in the visible range, and its easily observable electronic transitions, indigo is widely used as a model for studying the relationship between visible

absorption and molecular structure. In the experimental portion of this lab, the UV-Visible spectrum of indigo dissolved in ethanol was recorded and compared to the computed spectrum that was generated using density functional theory (DFT) and time-dependent DFT (TD-DFT). Comparing the experimental to the computational spectra allows for the observation of the accuracy of the computational models in simulating experimental results.

THEORY

Ultraviolet-visible (UV-Vis) spectroscopy utilizes the absorption of light by a molecule that corresponds to the electronic transition it undergoes. When a molecule absorbs a photon, an electron gets promoted from the highest occupied molecular orbital (HOMO) to the lowest unoccupied molecular orbital (LUMO). The energy difference between these orbitals determines the wavelength of the light absorbed. This can be mathematically expressed as

Equation 1: Planck's equation

$$E = h\nu = \frac{hc}{\lambda}$$

In this equation, h is Planck's constant, c is the speed of light, E is the energy of a photon, and λ is the wavelength of the absorbed light. Rearranging this equation results in an equation that can be used to calculate the wavelength corresponding to energy transitions.

Equation 2: Planck-Einstein relation

$$\lambda = \frac{hc}{E}$$

In UV-Vis spectroscopy, varying wavelengths of light are passed through a sample, and the raw transmittance of the molecule is measured. This value is used to calculate the amount absorbed at each wavelength. The result is a spectrum that plots the absorbance vs. the wavelength of the light, allowing for visualization with each peak corresponding to an electronic transition.

Computational spectra are based on quantum chemical methods, such as time-dependent density functional theory (TD-DFT), to calculate the electronic energy levels of a molecule and the transition energies between them. The result is a computational spectrum that can be plotted

against the UV-Vis spectrum for, in this case, comparison and validation.

Comparing the computational and experimental spectra allows for the theoretical model to be validated. If these peaks align closely with the experimental data for indigo, it allows for more certainty that the model accurately describes the molecule's electronic structure. If the result is vastly different, it reveals limitations of the model or inaccuracies in the UV-Vis experiment.

EXPERIMENTAL

Chemicals and Equipment

The dye indigo ($C_{16}H_{10}N_2O_2$) was used as the analyte, while ethanol was used as the solvent for preparing the blank and solution.

The UV-Visible absorption spectra as collected using a GENESYS 10S UV-Vis Spectrophotometer and the VisionLite software. Measurements were taken with quartz cuvettes, and the instrument was allowed to warm up for ten minutes before taking measurements.

The computational calculations were performed using Gaussian 16 on a Linux environment running the Red Hat distribution. GaussView 6 was used as the graphical interface. The MN15 density functional and 6-311G(**) basis set were used for all calculations.

Experimental Procedure

An indigo solution was prepared by dissolving the dye in ethanol until a uniform blue coloration was achieved. Another quartz cuvette filled with ethanol was used as a blank to zero the spectrophotometer. The indigo solution was then measured, and the spectrum was recorded using VisionLite in the 200-800nm range, covering both ultraviolet and visible regions.

Computational Methods

All computations were carried out using Gaussian 16 through GaussView 6 GUI on a Red Hat Linux workstation. A new directory was created, with all necessary files transferred to it before

beginning.

The ground-state geometry of indigo was optimized at the MN15-6-311G(**) level of theory. This was confirmed to be the optimized structure through the use of a harmonic vibrational frequency analysis, confirming that the structure was the true energy minimum. The first excited-state geometry of indigo was then optimized using time-dependent density functional theory (TD-DFT).

Vibrational frequency calculations were performed on both the ground and excited states so that a vibrationally resolved absorption spectrum could be generated. The Franck-Condon-Herzberg-Teller (FCHT) was utilized, resulting in the absorption spectrum.

The transition energies obtained from TD-DFT were converted into wavelengths using the Planck-Einstein relation equation. These calculated transitions were broadened using a Lorentzian function, which attempts to simulate realistic peak shapes, resulting in the final spectrum that can be used to compare with the experimental.

DATA

The electronic absorption spectra of the computational (Figure 1) and experimental methods used to analyze indigo (Figure 2) are displayed. Both figures show λ_{max} values at ~600-700nm.

Figure 3 contains an overlaid version of both spectra.

Figure 1: Computational Spectra of Indigo Dye Generated with Gaussian 16

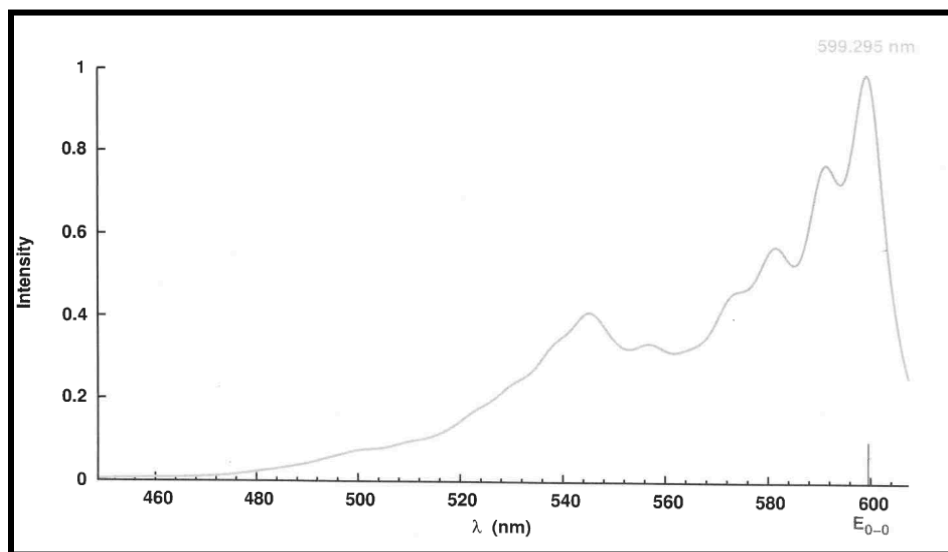
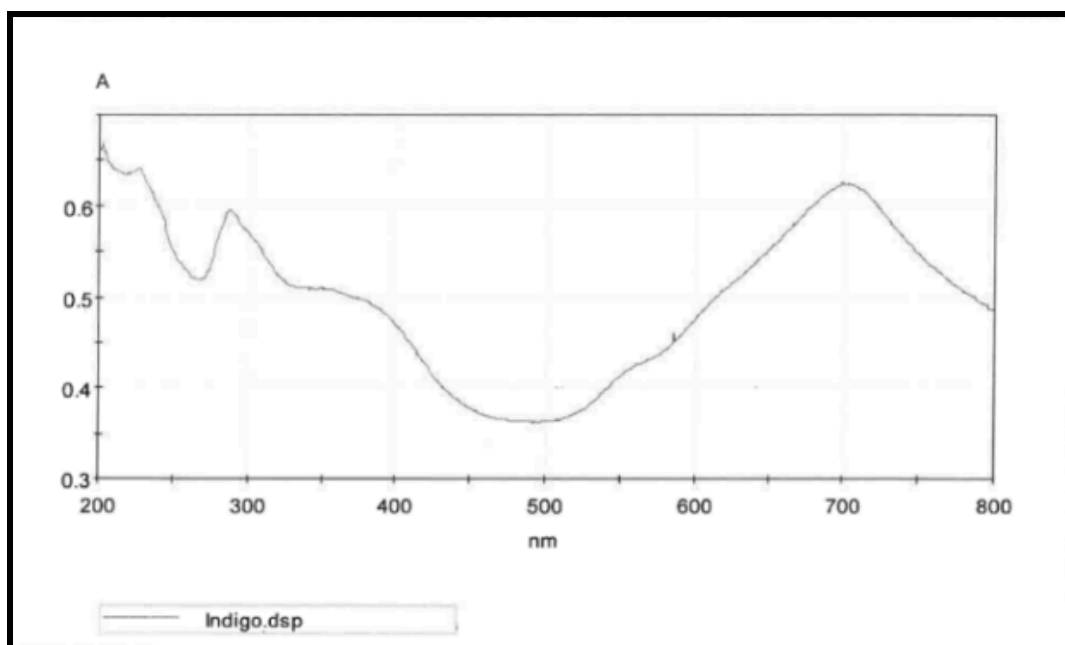
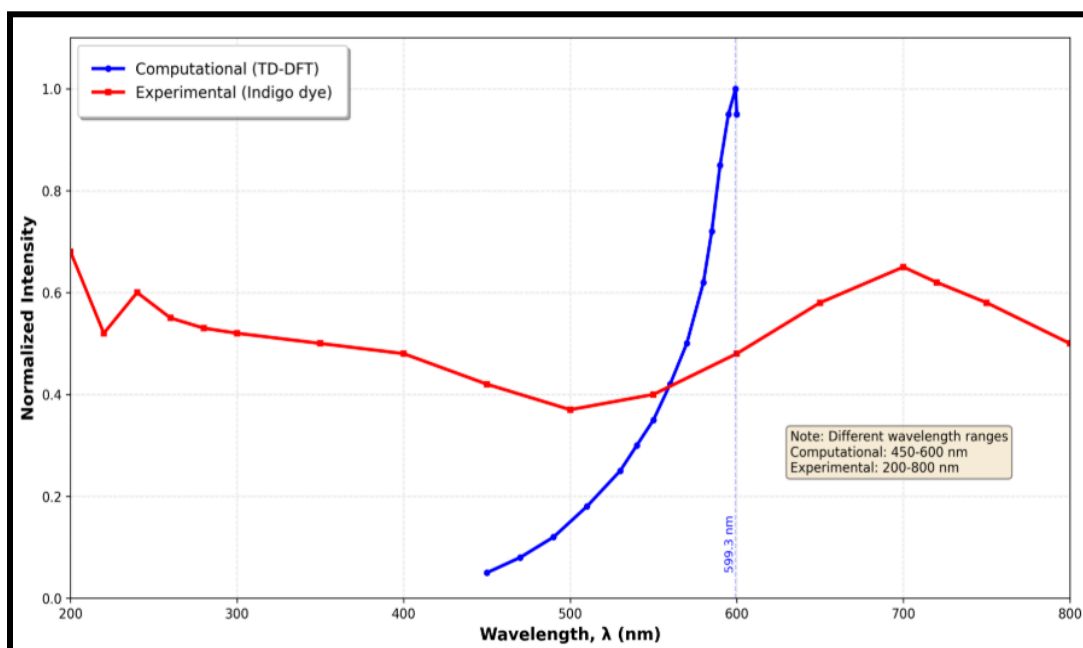


Figure 2: Experimental Spectra of Indigo Dye in Ethanol Solution**Figure 3:** Overlaid Computational and Experimental Spectra of Indigo Dye

RESULTS AND DISCUSSION

Analysis and Spectra Comparison

Figure 3 showcases an overlay of the computational and experimental UV-Visible absorption spectra for indigo. The computational spectrum was obtained at the MN15/6-311G(**) level using time-dependent density functional theory (TD-DFT) with vibrational structure included in the Franck-Condon-Herzberg-Teller approximation. This combination attempts to provide accurate excitation energies for organic chromophores without overly demanding computational cost.

In the overlaid spectra, the computational absorption maximum of indigo in ethanol is located at 590-600nm, which is significantly different than the experimental value that showcases indigo's maximum absorption at 700nm. This 100nm difference indicates that the MN15/6-311G(**) level underestimated the transition energy for the conjugated system. This could be due to solvent effects or potential aggregation effects in the experimental solution that aren't captured in the gas-phase calculation.

Despite the visible offset of the wavelengths, both spectra show absorption in the orange-red region of the visible spectrum, explaining indigo's characteristic blue appearance. A compound will show a color complementary to the light it absorbs, giving context to the 590-700nm results showing the reflection of blue light. The computational spectrum does predict the correct color that is consistent with the dye.

Physical Properties and Industrial Applications

Indigo's color is closely tied to the practical application of the dye. The dye exhibits low water solubility of approximately 0.99 mg/L at 25°C (Yuan, 2018), which gives context to why the dye had to be dissolved in ethanol rather than using water as the solvent. Indigo cannot be directly applied from water-based dye baths, making textile manufacturers first chemically reduce indigo to a soluble leuco-form, apply it to the fabric, then oxidize it back to the insoluble blue form in which it is trapped in the fiber matrix.

There is a large global demand for indigo as it continues its industrial importance. Approximately 70,000 metric tons of indigo (Linke, Rayat & Ward, 2023) are produced worldwide each year. Most of this indigo is chemically synthesized rather than naturally extracted. A pair of blue jeans typically requires 3-12 grams (Deakin & Cooper, 2018) of indigo dye to achieve a blue coloration. This showcases how dominant indigo is in the textile industry.

Aside from textiles, indigo is also widely utilized in food applications. Indigo Carmine (Blue No. 2) is a sulfonated form of indigo and was widely used in the United States until its ban in mid-2025. It was approved for beverages, processed foods, pharmaceuticals, and confectionery.

Conclusion

In summary, comparing the computational and experimental spectra in the orange-red region demonstrates strong agreement for indigo dye. While neither spectrum aligned perfectly, both clearly indicated the corresponding wavelengths, resulting in indigo's blue appearance. The MN15/6-331G(**) provides a reliable representation of indigo's electronic spectra. Indigo has a strong role in industrial and scientific importance.

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APPENDICES

Glossary:

Basis Set - A collection of mathematical functions to describe orbitals in quantum calculations.

Canonical orbitals - Delocalized molecular orbitals obtained from quantum chemistry calculations (ex., HOMO and LUMO)

CD command - A Linux command standing for change directory. Useful for navigating into different folders in the command line.

Checkpoint file (.chk) - A file type generated by Gaussian that stores the wavefunction as well as the optimized geometry of the molecule.

Density Functional Theory (DFT) - A Computational Quantum method that approximates electron correlation, which doesn't use wavefunctions, but uses electron density.

Electron Absorption Spectrum - A plot showing how a molecule absorbs light at different wavelengths.

GaussView - A graphical user interface (GUI) used to build molecules for Gaussian 09.

Gaussian 09 - A computational chemistry software for better analyzing quantum interactions.

Highest Occupied Molecular Orbital (HOMO) - The molecular orbital containing the highest-energy electrons in a given molecule.

Linux - An open-source operating system offering more customizability than alternatives like macOS or Windows.

Lowest Unoccupied Molecular Orbital (LUMO) - The lowest-energy molecular orbital that does not contain electrons.

Mkdir command - A Linux terminal command used to make a new directory (make a new folder).

Natural localized Molecular Orbitals (NLMOs) - Orbitals localized around specific bonds or atoms, rather than around canonical orbitals.

Scheme File - A file with a specified functional and base set for Gaussian calculations.

Time-Dependent Density Functional Theory (TD-DFT) - An extension of DFT to study electronically excited states of absorption spectra.

Wavefunction - A mathematical way that describes the behavior of a molecule's electrons.

Linux Shortcuts:

Cd - Change directory, used to navigate folders.

Cp - Copy command. Used to copy files to another location.

Ls - List command that displays the contents of a directory.

Tab Key - auto-completes when typing a command or file name

Up Arrow (↑) - Duplicates the previous command entered into the terminal.

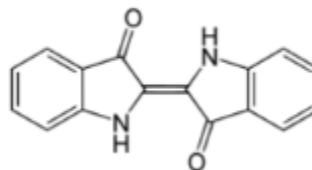
Down Arrow (↓) - Moves forward with command history after using the up arrow.

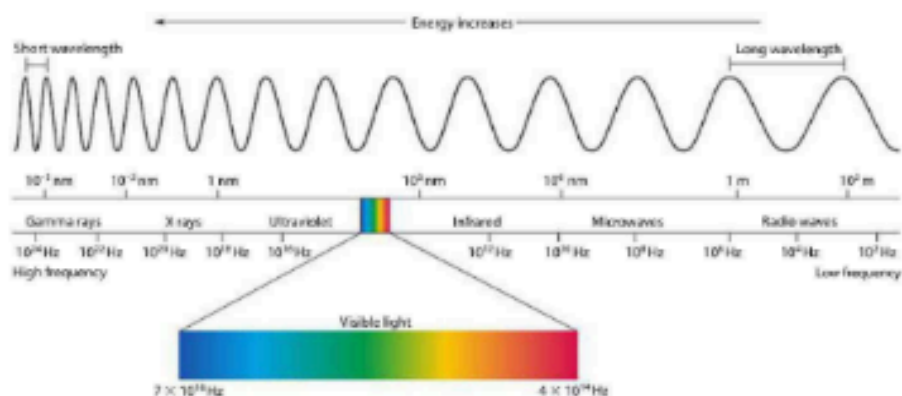
Pwd - Print working display, shows the full path of a directory.

Cat - Displays file contents.

Tail - Shows the last few lines of a file.

Indigo Dye (C₁₆H₁₀N₂O₂)





The electronic spectrum ranges from 200 to 800 nm, with the visible region being 400 to 800 nm. The region from 200 to 400 nm is the ultraviolet region of the spectrum. This range of electromagnetic radiation is representative of transitions between electronic energy states of a molecule. Each energy level is characterized by the degrees of freedom of the molecule. For the purpose of this experiment we will only concern ourselves with the electronic and vibrational degrees of freedom. A more complete discussion can be found in your text book. The transitions between energy levels of a molecule are represented by a Jablonski diagram

